Exact values

1 Zero

2

a $x = \frac{\pi}{4}$ **b** $x = \frac{\pi}{3}$ **c** $x = \frac{\pi}{4}$ **d** $x = \frac{\pi}{6}$

3

a $x = \frac{\pi}{3}, -\frac{2\pi}{3}$

b $x = \frac{\pi}{6}, -\frac{5\pi}{6}$

c $x = \frac{\pi}{12}, \frac{5\pi}{12}, -\frac{11\pi}{12}, -\frac{7\pi}{12}$

d $x = -\frac{\pi}{3}, \frac{2\pi}{3}$

4

a $\theta = 0, \pi, -\pi$

b $\theta = 0, \pi, 2\pi, 3\pi, 4\pi, -\pi, -2\pi, -3\pi, -4\pi$

c $\theta = \frac{\pi}{6}, -\frac{\pi}{6}$

d $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, -\frac{5\pi}{6}, -\frac{\pi}{6}$

e $\theta = 0, 2\pi, -2\pi$

f $\theta = \pm\frac{\pi}{6}, \pm\frac{5\pi}{6}, \pm\frac{7\pi}{6}, \pm\frac{11\pi}{6}, \pm\frac{13\pi}{6}, \pm\frac{17\pi}{6}, \pm\frac{19\pi}{6}, \pm\frac{23\pi}{6}$

5

a $x = \frac{\pi}{6}, \frac{5\pi}{6}$

b $x = \frac{5\pi}{6}, \frac{7\pi}{6}$

c $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

d $x = \frac{\pi}{4}, \frac{7\pi}{4}$

e $x = \frac{\pi}{4}, \frac{5\pi}{4}$

f $x = \frac{7\pi}{6}, \frac{11\pi}{6}$

g $x = \frac{\pi}{4}, \frac{5\pi}{4}$

h $x = \frac{3\pi}{4}, \frac{7\pi}{4}$

i $x = \frac{\pi}{12}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{11\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}$

j $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

k $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

l $x = \frac{3\pi}{2}$

m $x = \frac{2\pi}{3}$

n $x = \frac{\pi}{3}, \frac{5\pi}{3}$

o $x = 0$

p $x = \pi$

q $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

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6

a $x = \frac{\pi}{2}$

b $x = \pi$

c $x = \frac{\pi}{6}, \frac{5\pi}{6}$

d $x = \frac{5\pi}{4}, \frac{7\pi}{4}$

e $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

f $x = \frac{5\pi}{6}, \frac{11\pi}{6}$

g $x = \frac{\pi}{3}, \frac{2\pi}{3}$

h $x = \frac{\pi}{3}, \frac{5\pi}{3}$

i $x = \frac{\pi}{6}, \frac{7\pi}{6}$

j $x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$

k $x = \frac{5\pi}{18}, \frac{11\pi}{18}, \frac{17\pi}{18}, \frac{23\pi}{18}, \frac{29\pi}{18}, \frac{35\pi}{18}$

l $x = \frac{\pi}{12}, \frac{7\pi}{12}$

m $x = \frac{\pi}{2}, \frac{7\pi}{6}$

n $x = \frac{3\pi}{4}, \frac{7\pi}{4}$

o $x = \frac{11\pi}{6}, \frac{\pi}{2}$

p $x = \frac{4\pi}{3}$

q $x = \frac{\pi}{2}, \pi$

r $x = \frac{3\pi}{2}, \frac{\pi}{6}$

7

a $\theta = 0, 2\pi$

b $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

c $\theta = 0, \frac{2\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$

d $\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}$

Other radian values

8 Two solutions

9

a
- i $\theta = 0.35$
- ii $\theta = 0.35, 2.79$

b
- i $\theta = 0.44$
- ii $\theta = 0.44, 5.84$

c
- i $\theta = 0.70$
- ii $\theta = 0.70, 3.84$

d
- i $\theta = 1.33$
- ii $\theta = 1.33, 4.95$

e
- i $\theta = 1.45$
- ii $\theta = 1.45, 1.70$

f
- i $\theta = 0.66$
- ii $\theta = 0.66, 3.80$

g
- i $\theta = 0.84$
- ii $\theta = 0.84, 5.45$

h
- i $\theta = 0.09$
- ii $\theta = 0.09, 3.05$

i

i$\theta =$	ii$\theta =$
0.23	$0.23, 2.92, 3.37, 6.06$

10 $x = \pi - p$

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11 $x = 2\pi - n$

12 $x = \pi + t$

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